

Vruti Bharatbhai Patel

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SUMMARY

Master of Public Health (MPH) graduate with a concentration in Epidemiology and Biostatistics from Boston University, with hands-on experience in statistical programming (SAS, R), data management, and clinical trial operations. Proven ability to analyze large real-world datasets and apply epidemiological methods to maternal and child health research, global health studies, and observational trials. Former dentist with a passion for data-driven approaches to improving health outcomes. Seeking research-focused roles in public health, clinical research, or data analysis.

EDUCATION

Boston University School of Public Health

May 2025

Master of Public Health (MPH), Epidemiology and Biostatistics

- **GPA:** 3.87 / 4.0
- **Coursework:** Concepts and Methods of Epidemiology, Design and Conduct of Clinical Trials, Applied Statistics in Clinical Trials, Applied Intermediate Biostatistics, Cancer Epidemiology, Public Health Law and Policy

Karnavati School of Dentistry - Gujarat University

Oct 2021

Bachelor of Dental Surgery (BDS)

- **GPA:** 3.56 / 4.0

SKILLS

- **Programming Languages:** SAS, R
- **Data Management:** Data Analysis (Multivariable Linear and Logistic Regression, Survival Analysis, Cox Proportional Hazard Regression), Data Entry, eCRF Management, Quantitative and Qualitative Data Collection
- **Technical Skills:** Data Visualization, Data Validation, Microsoft Word, Advanced Excel, REDCap, Medidata RAVE
- **Research Skills:** Comprehensive Literature Review, Critical Evaluation, IRB Applications, Clinical Trial Management
- **Communication:** Interpersonal Communication, Community Engagement

EXPERIENCE

Department of Global Health - Boston University School of Public Health

Oct 2024 - May 2025

Research Intern

Boston, MA

- Managed and cleaned large real-world MNHR datasets (~5,000-10,000 records) from multiple study sites in India, performing quality checks, validation procedures, and resolving data discrepancies to ensure accuracy and integrity.
- Conducted descriptive and inferential statistical analyses using SAS and R to examine maternal and neonatal health outcomes, including trends in preterm birth, neonatal mortality, and gestational complications.
- Applied epidemiological methods to support study objectives, including variable derivation, data visualization, and model building for exploratory analysis.
- Synthesized findings from 15+ peer-reviewed articles to inform data interpretation and support the development of research reports, abstracts, and presentations for internal stakeholders.
- Collaborated with interdisciplinary research teams to translate analytical results into actionable recommendations aimed at improving maternal and newborn health care practices in low-resource settings.

Boston University School of Public Health

Sep 2024 - May 2025

Graduate Teaching Assistant – Epidemiology and Biostatistics

Boston, MA

- Led weekly discussion sections for 40+ graduate students, reinforcing epidemiological concepts, study design, and biostatistical methods to support course objectives.
- Provided detailed grading for 100+ assignments and exams, offering timely, constructive feedback that improved student performance and understanding.
- Collaborated with faculty to develop instructional materials and facilitated communication between students and instructors, enhancing course delivery and engagement for a diverse cohort.

Boston Medical Centre - Boston University School Of Medicine

Dec 2023 - Aug 2024

Student Research Assistant

Boston, MA

- Collaborated with research teams to recruit and screen eligible study participants using EPIC electronic health records (EHR), evaluating patient data to ensure protocol eligibility and regulatory compliance according to Good Clinical Practice (GCP) guidelines.
- Managed IRB submissions and correspondence, maintaining strict adherence to ethical standards and regulatory requirements across multiple study protocols.
- Supported clinical trial operations by updating case report forms (CRFs), resolving data entry queries, and creating standardized study visit checklists to optimize data quality and protocol compliance.
- Maintained and organized essential trial documents, including clinical trial master files and safety reports, ensuring audit readiness and regulatory documentation integrity.

- Facilitated participant engagement through consistent communication, ensuring high levels of retention, accountability, and compliance during data collection and study visits.

Sumukh Dental Care

Associate Dentist

Nov 2021 - Jun 2023

Ahmedabad, India

- Diagnosed and steered new patients toward the best treatment option for their oral health issues
- Provided patients with optimal oral health and well-being through scaling, restoration, root canal treatments, and pharmacologic interventions for prompt and effective pain relief.
- Supervised patient schedules, optimized appointment efficiency, and kept a case library for continuous improvement

ACADEMIC PROJECTS

Digoxin Investigation Group Trial-Data Analysis

- Developed Survival Analysis and Cox-Proportional Hazard models to investigate the association between Left Ventricular Ejection Fraction and Stroke using DIG Trial data

Turmeric, Tea and COVID-19

- Designed a hypothetical comprehensive clinical trial protocol along with detailed consent form in a team of four.

Behavioral Risk Factor Surveillance System (BRFSS)-Data Analysis

- Used ANOVA and multivariate regression analysis to identify the association between sexual orientation and mental health in BRFSS dataset using SAS

Framingham Heart Study-Data Analysis

- Statistically analyzed and wrote a report on a limited data set to test the relationship between high cholesterol levels and cardiovascular disease occurrence using R programming language

US COVID-19 State Policies Dashboard

- Cleaned and analyzed the US COVID-19 State Policies Dataset to create visual representation of vaccine distribution policies among incarcerated individuals in US States using Microsoft Excel

Pesticide Exposure and Non-Hodgkin's Lymphoma-A Literature Review

- Conducted an extensive review to assess the impact of pesticide exposure on the risk of onset of Non-Hodgkin's Lymphoma

ACTIVITIES AND COMMUNITY SERVICE

Special Olympics Bharat Camp

Volunteer

- Volunteered in Special Olympics Bharat Camp to diagnose intellectually disabled children for full body health screenings

Karnavati School of Dentistry

Volunteer

- Effectively taught oral hygiene habits to over 50 students at rural schools during dental camps organized by Karnavati School of Dentistry

CITI CERTIFICATIONS

- HIPAA and Data Security certified
- CITI certified for Good Clinical Practice and Biomedical Research